

RAZA SHAUD

Electrical Engineer | Machine Learning & Computer Vision | Embedded Systems & IoT

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PROFESSIONAL SUMMARY

Electrical Engineering graduate (B.Tech, 2026) and GATE 2026 qualifier (AIR 2139) with hands-on experience across machine learning, computer vision, embedded systems, and power engineering. Delivered 4 projects achieving up to 96.5% model accuracy and 99% process automation; completed 3 internships spanning ML development, steel manufacturing (SAIL), and thermal power generation (DVC). Published 3 research papers, including an IEEE Best Paper Award on PMSM motor design for electric vehicles.

TECHNICAL SKILLS

- Programming:** Python, C, C++
- Machine Learning:** Deep Learning, CNN, Random Forest, SVM, Feature Engineering, Data Preprocessing, Scikit-learn, NumPy, Pandas
- Computer Vision & Signal Processing:** OpenCV, YOLOv5, Image Processing, MFCC Feature Extraction
- Embedded Systems & IoT:** Arduino (Uno), Microcontroller Programming, Sensor Interfacing, GSM Communication, Real-Time Monitoring, Hardware Prototyping
- Electrical & Power:** ETAP, PMSM, Industrial Automation, MATLAB, ANSYS
- Tools:** Git, Arduino IDE, Microsoft Office (Excel, Word, PowerPoint)

PROFESSIONAL EXPERIENCE

Machine Learning Intern | Tech Solutions Inc. | Kolkata, West Bengal **May 2025 – Jul 2025**

- Automated Python-based data preprocessing pipelines for 5,000+ records, cutting data preparation time by 60%.
- Optimized 8 machine learning scripts, improving model prediction speed by 35%.
- Tested and validated classification models with a 4-person team, achieving 92% accuracy across 3 business use cases.

Internship Trainee | Steel Authority of India Limited (SAIL) | Bokaro, Jharkhand **Nov 2024 – Dec 2024**

- Studied industrial automation processes and plant electrical systems in a large-scale steel manufacturing environment.

Internship Trainee | Damodar Valley Corporation (DVC) | Mejia, West Bengal **Jun 2024 – Jul 2024**

- Shadowed senior engineers at a major thermal power generation plant, documenting daily operations and plant electrical systems.

PROJECTS

Solar-Powered Robotic Cable Monitoring System | Python, Deep Learning (CNN), OpenCV, Arduino, Solar PV

- Engineered an autonomous robot integrating a 12V solar PV system with camera-based cable insulation monitoring across 500m of infrastructure, eliminating 100% of manual inspection.
- Built a CNN fault-detection model on 1,300+ cable images achieving 96.5% accuracy, cutting false positives by 78% vs. traditional visual inspection.
- Optimized power draw for 8-hour continuous operation on a 20W solar panel, saving ₹15,000 per unit annually.

Human vs AI Voice Classification System | Python, Scikit-learn, Random Forest, MFCC, Audio Signal Processing

- Developed a binary classifier on 1,500+ audio samples using MFCC feature extraction and spectral-gating noise reduction (65% less interference), achieving 89% accuracy.
- Implemented Random Forest with a 15-dimensional feature vector, outperforming a baseline SVM by 12% via hyperparameter tuning and 5-fold cross-validation.

Face Recognition Attendance System | Python, OpenCV, Pandas, Computer Vision

- Built a real-time facial recognition system processing 30 faces/second, automating attendance for 100+ students and reducing manual record-keeping by 99%.
- Achieved 95% recognition accuracy across 200+ daily scans under varying lighting, cutting processing from 15 minutes to 45 seconds per session (28 instructor-hours saved per semester).

Saline Monitoring and Alert System | Arduino Uno, Embedded C, Ultrasonic Sensor, GSM Module, IoT

- Designed an IoT medical monitoring system tracking saline levels with 2cm precision, preventing 100% of empty-bottle incidents across 50 hours of testing.
- Programmed dual alerts (buzzer + GSM SMS to 3 medical staff contacts), reducing response time by 75% and nursing workload by 40% in clinical simulations.

EDUCATION

Bachelor of Technology (B.Tech), Electrical Engineering

Nov 2022 – Jul 2026

Ghani Khan Choudhury Institute of Engineering and Technology, Malda, West Bengal

PUBLICATIONS

- "Characteristics Investigation of Dual Rotor Single Stator 6-Phase Single I-Shaped Magnetic Pole PMSM for Smart Electric Vehicles" — IEEE GIST 2024 International Conference (**Best Paper Award**)
- "Noise Filtration and Voice Discrimination: Case Studies, Results, and Discussions" — Tech Research Chronicles (IAST), Vol. 1, Issue 2, 2025, ISSN: 3049-205X
- "Intelligent Remote Monitoring of Low Voltage Cable Insulation Fault" — PEIS-2026, GKCIET, Malda, Apr 2026

AWARDS & CERTIFICATIONS

- GATE 2026 Qualified — Electrical Engineering (EE):** Score 515, All India Rank 2139
- 100 Days of Code: The Complete Python Pro Bootcamp** — Udemy, 2024